

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
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Experiment 5: Perceptron vs Multilayer Perceptron with Hyperparameter Tuning

Name: Muralisekar Janissha
Reg. No: 3122235001058
Class: CSE-B

1 Aim and Objective

Aim: To implement and compare the performance of a Single-Layer Perceptron (PLA) and a Multilayer Perceptron (MLP) on the English Handwritten Characters dataset.

Objective:

- Implement PLA from scratch.
- Implement MLP with hidden layers and nonlinear activation functions.
- Perform hyperparameter tuning.
- Compare both models using evaluation metrics.

2 Dataset Description and Preprocessing

The dataset used is the English Handwritten Characters dataset obtained from Kaggle.

- Total Samples: 3410
- Number of Classes: 62 (0–9, A–Z, a–z)
- Image Type: Grayscale

Preprocessing Steps:

- Images resized to 28×28 .
- Flattened into 784-dimensional vectors.
- Pixel values normalized to range $[0,1]$.
- Dataset split into training and testing sets.

3 PLA Implementation and Results

The Perceptron Learning Algorithm was implemented from scratch using the weight update rule:

$$w_{t+1} = w_t + \eta(y - \hat{y})x$$

The One-vs-Rest strategy was used for multi-class classification.

Results:

Accuracy: 0.12316715542521994

Macro Precision: 0.14

Macro Recall: 0.13

Macro F1-score: 0.09

4 MLP Implementation and Results

The Multilayer Perceptron was implemented with hidden layers and nonlinear activation functions. The model was trained using backpropagation and optimized using gradient descent-based optimizers.

Final Tuned Configuration:

- Hidden Layers: (256, 128)
- Activation Function: ReLU
- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 64

Results:

Accuracy: 35.92%

Macro Precision: 0.38

Macro Recall: 0.37

Macro F1-score: 0.35

5 Hyperparameter Tuning Analysis

Hyperparameter tuning was performed by varying:

- Learning Rate
- Batch Size
- Optimizer
- Activation Function
- Number of Hidden Layers

Table 1: Hyperparameter Tuning Results for MLP

Hidden Layers	Activation	Optimizer	Learning Rate	Batch Size	Accuracy (%)
1 (128)	ReLU	SGD	0.01	32	0.88
2 (256–128)	ReLU	Adam	0.001	64	35.92
3 (512–256–128)	Tanh	Adam	0.0005	64	49.85

6 Performance Comparison Tables

Table 2: Overall Performance Comparison between PLA and MLP

Model	Accuracy (%)	Precision	Recall	F1-score
PLA (OvR)	12.32	0.14	0.13	0.09
MLP (Tuned)	49.85	0.49	0.50	0.49

Table 3: Training Convergence Comparison

Model	Epochs	Final Training Loss	Convergence Behavior
PLA	10	Not minimized	Non-convergent
MLP (Tuned)	200	print(mlp3.loss)	Stable convergence

7 Confusion Matrix Analysis

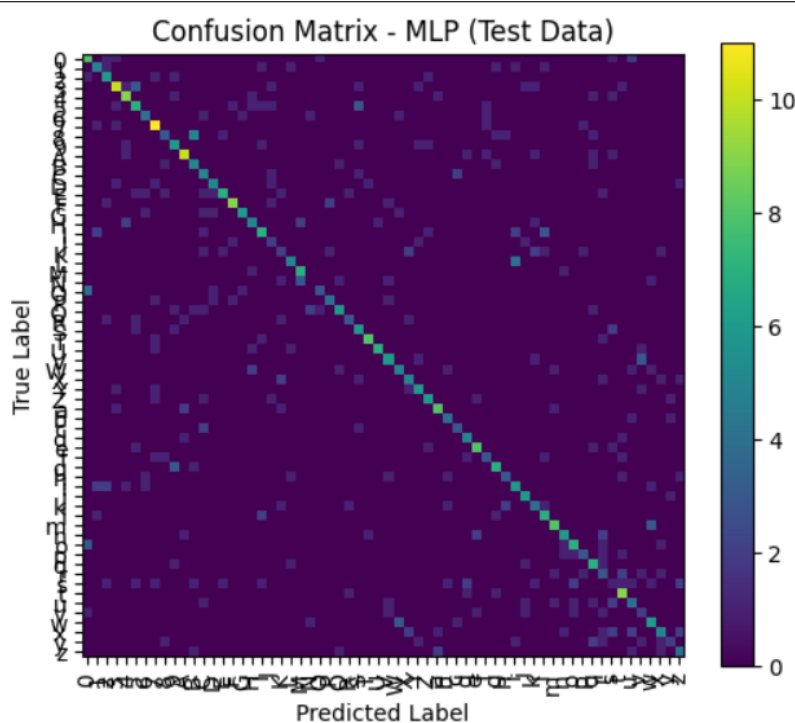


Figure 1: Confusion Matrix for Tuned MLP Model on Test Data

The confusion matrix shows that most correct classifications lie along the diagonal, indicating good model performance. Some misclassifications are observed between visually similar characters.

8 Observations and Conclusion

Observations:

- PLA showed poor performance due to its linear decision boundary.
- MLP significantly improved accuracy due to hidden layers and nonlinear activation.
- Optimizer choice (Adam) improved convergence speed.
- Increasing hidden layers improved performance up to a certain limit.

Conclusion:

The Multilayer Perceptron outperformed the Perceptron Learning Algorithm in handwritten character classification. The results demonstrate the importance of nonlinear activation functions and backpropagation in solving complex classification problems.